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Unit – 5 : Session – 6 : SLO – 1 | ARM Instruction Set Architecture

■ I. Multiple Choice

1. b) Reduced Instruction Set Computing (RISC).
2. b) Thumb — 16-bit instructions reduce code size and improve memory efficiency.
3. c) Multiple load/store (LDM/STM) — transfer blocks of registers in one instruction.
4. c) Change the program counter to a target address.
5. c) NEON — ARM's SIMD extension.

■ II. True or False

1. TRUE.
2. FALSE — Thumb is designed to decrease code size.
3. TRUE.
4. TRUE.
5. TRUE.

■ III. Scenario-Based Activity

1. Thumb — for memory-constrained parts, its 16-bit instructions give much better code density than 32-bit ARM code.
2. LDM/STM (Load/Store Multiple) — move whole register sets between memory and the register file in a single instruction, cutting instruction count for the repeated bulk transfers.
3. NEON — SIMD processes multiple audio/video samples per instruction (e.g. 8×16-bit values at once), accelerating the vector-heavy computation loops and keeping the streams real-time.